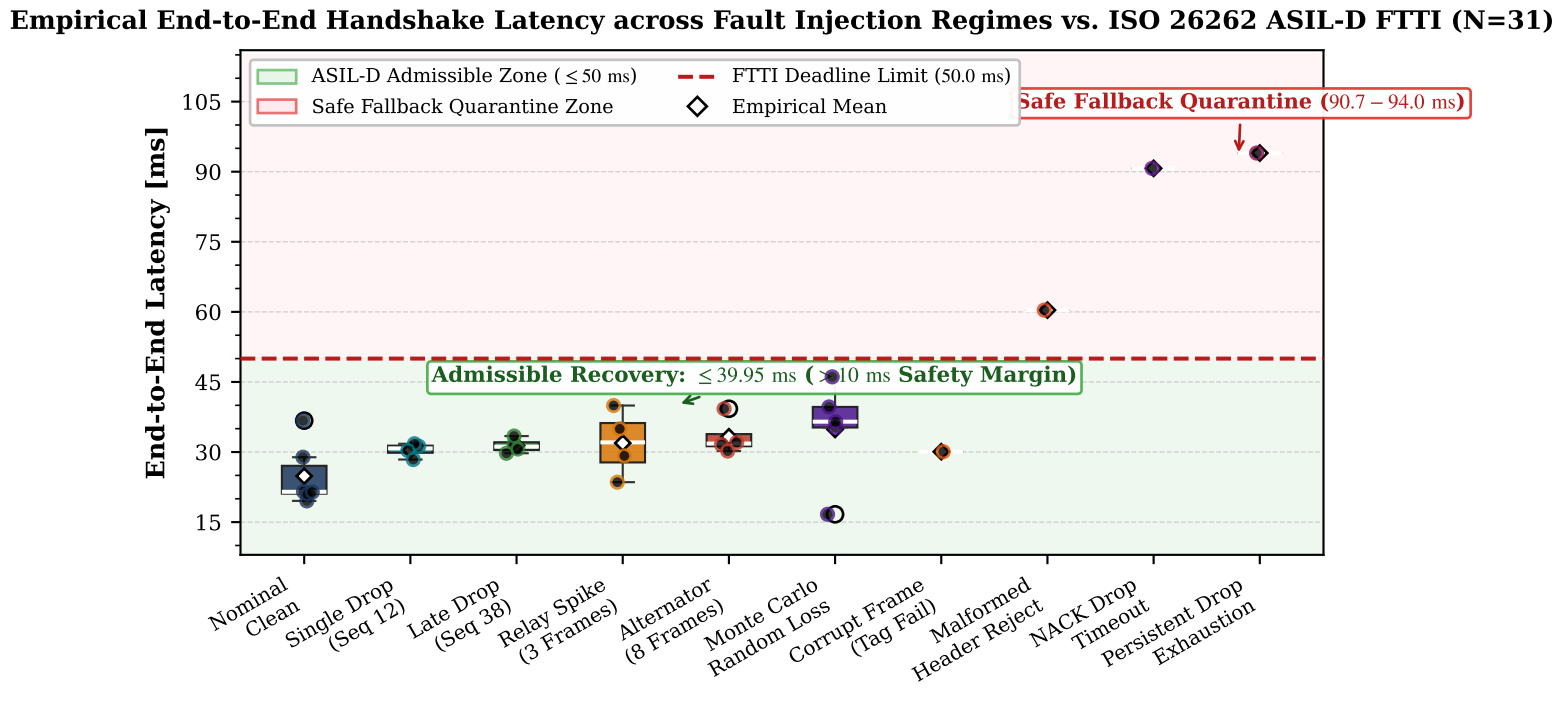
Empirical End-to-End Handshake Latency across Fault Injection Regimes vs. ISO 26262 ASIL-D FTTI (N=31)
ASIL-D Admissible Zone (≤ 50 ms)
Safe Fallback Quarantine Zone
FTTI Deadline Limit (50.0 ms)
Empirical Mean
Safe Fallback Quarantine (90.7 – 94.0 ms)
Admissible Recovery: ≤ 39.95 ms (≥ 10 ms Safety Margin)
End-to-End Latency [ms]
105
90
75
60
45
30
15
Nominal Clean
Single Drop (Seq 12)
Late Drop (Seq 38)
Relay Spike (3 Frames)
Alternator (8 Frames)
Monte Carlo Random Loss
Corrupt Frame (Tag Fail)
Malformed Header Reject
NACK Drop Timeout
Persistent Drop Exhaustion